

Problem

Find the Laplace transform of $f(t) = (\cos(2t) + e^{-4t})u(t)$.

Step-by-step solution

Step 1 of 2

The signal $f(t) = (\cos 2t + e^{-4t})u(t)$ and is defined as,

$$\begin{aligned} f(t) &= \cos 2t + e^{-4t} \quad \text{when } u(t) = 1 \text{ for } t \geq 0, \\ &= 0 \quad \text{when } u(t) = 0 \text{ for } t < 0. \end{aligned}$$

Write $\cos 2t$ signal in terms of exponential signal to obtain easy calculation.

$$\cos 2t = \frac{e^{j2t} + e^{-j2t}}{2}$$

Use linearity property for evaluating the Laplace transform for the signal $f(t)$.

$$\begin{aligned} \mathcal{L}(f(t)) &= \mathcal{L}\left(\frac{e^{j2t} + e^{-j2t}}{2} + e^{-4t}\right)u(t) \\ &= \frac{1}{2}\mathcal{L}(e^{j2t})u(t) + \frac{1}{2}\mathcal{L}(e^{-j2t})u(t) + \mathcal{L}(e^{-4t}u(t)) \\ &= \frac{1}{2}\int_0^\infty e^{j2t}e^{-st}dt + \frac{1}{2}\int_0^\infty e^{-j2t}e^{-st}dt + \int_0^\infty e^{-4t}e^{-st}dt \\ &= \frac{1}{2}\int_0^\infty e^{-(s-j2)t}dt + \frac{1}{2}\int_0^\infty e^{-(s+j2)t}dt + \int_0^\infty e^{-(s+4)t}dt \end{aligned}$$

Step 2 of 2

Evaluate the integrals.

$$\begin{aligned}
 \mathcal{L}(f(t)) &= \frac{1}{2} \frac{e^{-(s-j2)t}}{-(s-j2)} \Big|_0^\infty + \frac{1}{2} \frac{e^{-(s+j2)t}}{-(s+j2)} \Big|_0^\infty + \frac{e^{-(s+4)t}}{-(s+4)} \Big|_0^\infty \\
 &= \left[-\frac{1}{2(s-j2)} \left(e^{-(s-j2)\infty} - e^{-(s-j2)0} \right) \right. \\
 &\quad \left. -\frac{1}{2(s+j2)} \left(e^{-(s+j2)\infty} - e^{-(s+j2)0} \right) \right. \\
 &\quad \left. -\frac{1}{(s+4)} \left(e^{-(s+4)\infty} - e^{-(s+4)0} \right) \right] \\
 &= \left[-\frac{1}{2(s-j2)}(0-1) - \frac{1}{2(s+j2)}(0-1) - \frac{1}{(s+4)}(0-1) \right] \\
 &= \frac{1}{2(s-j2)} + \frac{1}{2(s+j2)} + \frac{1}{(s+4)} \\
 &= \frac{1}{2} \left(\frac{2s}{s^2+4} \right) + \frac{1}{(s+4)} \\
 &= \frac{s}{s^2+2^2} + \frac{1}{s+4} \\
 &= \frac{s(s+4) + s^2+4}{(s+4)(s^2+4)} \\
 &= \frac{2s^2+4s+4}{(s+4)(s^2+4)}
 \end{aligned}$$

Therefore, the Laplace transform of the signal $(\cos 2t + e^{-4t})u(t)$ is $\boxed{\frac{2s^2+4s+4}{(s+4)(s^2+4)}}$.